

## ROHRBACH, Austria

WMO: 110080

Lat: 48.570N

Lon: 13.994E

Elev: 1976

StdP: 13.68

Time Zone: 1.00 (EUC)

Period: 97-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 10.5       | 14.8 | 5.7                           | 7.9 | 12.3 | 10.1 | 9.9 | 17.0 | 15.8                  | 37.5 | 13.9 | 36.1 | 2.6                   | 20   | 0.489 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 16.5                   | 84.7            | 67.5 | 81.3 | 66.4 | 78.2 | 64.6 | 69.0                | 81.4 | 67.3 | 78.8 | 65.7 | 76.0 | 4.8                  | 190  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 64.7                            | 98.5 | 74.1 | 63.1 | 93.1 | 71.9 | 61.8 | 88.9 | 70.6 | 34.3          | 81.5 | 32.8 | 79.1 | 31.6 | 76.1 | 77.2              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |     |                    |     | n-Year Return Period Values of Extreme Temperature |     |            |      |            |      |            |      |      |
|-------------------|------|-----|----------------------------|-----|--------------------|-----|----------------------------------------------------|-----|------------|------|------------|------|------------|------|------|
| 1%                | 2.5% | 5%  | Mean                       |     | Standard Deviation |     | n=5 years                                          |     | n=10 years |      | n=20 years |      | n=50 years |      |      |
|                   |      |     | Min                        | Max | Min                | Max | Min                                                | Max | Min        | Max  | Min        | Max  | Min        | Max  |      |
| (a)               | (b)  | (c) | (e)                        | (f) | (g)                | (h) | (i)                                                | (j) | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |      |
| 12.3              | 10.9 | 9.5 | DB                         | 5.7 | 89.7               | 5.9 | 2.4                                                | 1.5 | 91.4       | -2.0 | 92.9       | -5.3 | 94.2       | -9.6 | 96.0 |
|                   |      |     | WB                         | 5.1 | 72.2               | 5.4 | 1.9                                                | 1.2 | 73.5       | -1.9 | 74.6       | -5.0 | 75.6       | -8.9 | 77.0 |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 47.4   | 28.6  | 31.4  | 38.5  | 48.4  | 55.8  | 62.1  | 64.6  | 64.5  | 56.2  | 47.7  | 37.9  | 31.5  |
|                                                                        | DBStd       | 14.62  | 7.62  | 7.55  | 7.43  | 7.77  | 7.03  | 6.99  | 6.24  | 6.19  | 6.16  | 6.72  | 6.84  | 6.77  |
|                                                                        | HDD50       | 2767   | 663   | 520   | 359   | 120   | 26    | 3     | 0     | 0     | 12    | 124   | 365   | 575   |
|                                                                        | HDD65       | 6650   | 1128  | 940   | 822   | 500   | 295   | 135   | 86    | 86    | 270   | 536   | 812   | 1040  |
|                                                                        | CDD50       | 1802   | 0     | 0     | 2     | 71    | 205   | 367   | 453   | 449   | 199   | 53    | 3     | 0     |
|                                                                        | CDD65       | 210    | 0     | 0     | 0     | 1     | 9     | 49    | 74    | 70    | 7     | 0     | 0     | 0     |
|                                                                        | CDH74       | 1810   | 0     | 0     | 0     | 10    | 99    | 436   | 618   | 592   | 55    | 0     | 0     | 0     |
|                                                                        | CDH80       | 418    | 0     | 0     | 0     | 0     | 11    | 100   | 152   | 149   | 6     | 0     | 0     | 0     |
| (14) Wind                                                              | WSAvg       | 3.9    | 4.2   | 4.3   | 4.3   | 4.4   | 4.3   | 4.0   | 3.9   | 3.6   | 3.5   | 3.8   | 3.4   | 3.7   |
| (15) Precipitation                                                     | PrecAvg     | 40.6   | 3.3   | 3.0   | 3.3   | 2.3   | 3.7   | 4.0   | 4.7   | 4.2   | 3.1   | 2.9   | 2.9   | 3.3   |
|                                                                        | PrecMax     | 60.6   | 7.8   | 7.0   | 8.6   | 4.3   | 6.7   | 7.9   | 9.9   | 15.1  | 7.2   | 7.2   | 7.0   | 7.7   |
|                                                                        | PrecMin     | 29.8   | 0.4   | 0.5   | 0.7   | 0.2   | 0.6   | 1.9   | 1.2   | 1.0   | 1.0   | 0.6   | 0.0   | 1.1   |
|                                                                        | PrecStd     | 6.5    | 2.0   | 1.7   | 2.2   | 1.0   | 1.7   | 1.5   | 2.1   | 2.6   | 1.5   | 1.9   | 1.6   | 1.7   |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 49.1  | 52.9  | 62.8  | 75.1  | 81.4  | 87.0  | 88.6  | 88.9  | 79.1  | 70.6  | 59.8  | 49.2  |
|                                                                        |             | MCWB   | 44.8  | 43.5  | 49.9  | 57.5  | 65.1  | 68.7  | 69.3  | 67.5  | 64.1  | 58.2  | 51.4  | 45.3  |
|                                                                        | 2%          | DB     | 44.7  | 48.1  | 58.6  | 69.9  | 76.8  | 83.2  | 84.1  | 84.1  | 75.0  | 65.4  | 55.1  | 45.7  |
|                                                                        |             | MCWB   | 41.6  | 42.6  | 48.3  | 55.7  | 62.2  | 67.4  | 67.7  | 66.8  | 62.8  | 56.2  | 49.3  | 43.2  |
|                                                                        | 5%          | DB     | 41.5  | 45.0  | 54.2  | 66.0  | 73.0  | 79.2  | 80.5  | 80.2  | 70.9  | 61.3  | 51.2  | 43.1  |
|                                                                        |             | MCWB   | 39.2  | 40.5  | 45.7  | 53.5  | 60.2  | 65.0  | 66.4  | 65.9  | 60.4  | 54.7  | 47.3  | 40.8  |
|                                                                        | 10%         | DB     | 38.5  | 41.6  | 50.1  | 62.3  | 69.1  | 75.4  | 77.1  | 76.8  | 67.4  | 57.9  | 47.8  | 40.3  |
|                                                                        |             | MCWB   | 36.7  | 38.0  | 44.1  | 51.6  | 57.9  | 63.2  | 64.7  | 64.4  | 59.1  | 53.1  | 45.0  | 38.4  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 45.9  | 45.7  | 51.2  | 59.1  | 66.6  | 70.6  | 71.8  | 70.8  | 65.9  | 59.9  | 52.5  | 46.5  |
|                                                                        |             | MCDB   | 48.6  | 50.2  | 61.1  | 72.9  | 78.2  | 84.1  | 84.4  | 83.2  | 75.9  | 67.0  | 57.7  | 48.4  |
|                                                                        | 2%          | WB     | 42.0  | 43.1  | 48.9  | 56.4  | 63.4  | 68.0  | 69.1  | 68.3  | 63.5  | 57.5  | 50.0  | 43.8  |
|                                                                        |             | MCDB   | 44.3  | 47.3  | 56.5  | 68.4  | 74.4  | 80.3  | 81.2  | 80.6  | 73.1  | 63.4  | 54.3  | 45.6  |
|                                                                        | 5%          | WB     | 39.4  | 40.7  | 46.8  | 54.3  | 61.1  | 66.1  | 67.2  | 66.7  | 61.5  | 55.4  | 47.6  | 41.0  |
|                                                                        |             | MCDB   | 41.4  | 44.6  | 52.8  | 64.7  | 71.3  | 76.9  | 78.5  | 78.0  | 69.5  | 60.0  | 50.9  | 42.9  |
|                                                                        | 10%         | WB     | 36.9  | 38.3  | 44.6  | 52.2  | 58.9  | 64.2  | 65.6  | 65.0  | 59.8  | 53.6  | 45.0  | 38.6  |
|                                                                        |             | MCDB   | 38.4  | 41.3  | 49.3  | 61.3  | 67.6  | 73.5  | 75.2  | 75.3  | 66.1  | 57.8  | 47.6  | 40.2  |
| (35) Mean Daily Temperature Range                                      | MDBR        |        | 8.5   | 10.6  | 13.5  | 16.6  | 16.2  | 16.6  | 16.5  | 16.8  | 15.0  | 13.4  | 9.7   | 8.2   |
|                                                                        | 5% DB       | MCDBR  | 11.0  | 16.7  | 22.3  | 23.2  | 23.8  | 23.2  | 22.9  | 23.4  | 20.8  | 19.5  | 16.5  | 12.1  |
|                                                                        |             | MCWBR  | 8.5   | 11.3  | 13.2  | 12.0  | 11.2  | 10.1  | 9.7   | 9.9   | 10.3  | 11.7  | 11.1  | 9.6   |
|                                                                        | 5% WB       | MCDBR  | 10.4  | 15.6  | 20.1  | 22.0  | 21.6  | 21.3  | 20.9  | 21.6  | 19.1  | 17.6  | 15.1  | 11.7  |
|                                                                        |             | MCWBR  | 8.4   | 10.9  | 12.4  | 11.7  | 10.6  | 9.8   | 9.3   | 9.8   | 10.2  | 11.3  | 10.6  | 9.7   |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.288 | 0.312 | 0.352 | 0.386 | 0.392 | 0.394 | 0.403 | 0.396 | 0.370 | 0.341 | 0.315 | 0.288 |
|                                                                        | taud        |        | 2.394 | 2.358 | 2.320 | 2.277 | 2.296 | 2.319 | 2.294 | 2.328 | 2.374 | 2.440 | 2.466 | 2.447 |
|                                                                        | Ebn at Noon |        | 248   | 267   | 272   | 273   | 276   | 275   | 271   | 267   | 263   | 253   | 236   | 234   |
|                                                                        | Edh at Noon |        | 21    | 27    | 34    | 39    | 40    | 39    | 40    | 37    | 32    | 25    | 20    | 18    |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 315   | 602   | 939   | 1369  | 1590  | 1744  | 1682  | 1487  | 1060  | 659   | 334   | 243   |
|                                                                        | RadStd      |        | 52    | 81    | 119   | 172   | 187   | 153   | 161   | 151   | 139   | 75    | 37    | 27    |

## Historical Trends

|                              | DBAvg | Heating |     |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------------|-------|---------|-----|--------|---------|-------|-------|-------------|-------|-------|-------|
|                              |       | 99% DB  |     | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                              |       | (d)     | (e) | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| (46) Station Only            | N/A   | N/A     | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (97 neighbors) | +0.90 | N/A     | N/A | N/A    | N/A     | +0.72 | +0.65 | N/A         | N/A   | +119  | +50   |

Nomenclature: See separate page